

ASTHMA AND COPD: DIAGNOSIS

Asthma and chronic obstructive pulmonary disease (COPD) both present with cough, wheeze, tight chest or difficulty breathing. Distinguish asthma from COPD:

Asthma likely if several of:

- Onset before 20 years of age
- Associated hayfever, allergic conjunctivitis or eczema, other allergies
- Intermittent symptoms with normal breathing in between
- Symptoms worse at night, early morning, with cold or stress
- Patient or family have a history of asthma
- PEFR¹ response to inhaled beta-agonist (e.g. salbutamol) improves $\geq 20\%$ (see below).

Give routine asthma care → 125.

COPD likely if several of:

- Onset after 40 years of age
- Symptoms are persistent and worsen slowly over time
- Cough with sputum starts long before difficulty breathing
- History of heavy smoking or worked in dusty environment
- Previous diagnosis of TB
- Previous doctor diagnosis of COPD

Give routine COPD care → 126.

Doctor to confirm diagnosis. If doctor not available, treat as asthma → 125 and refer to doctor within 1 month.

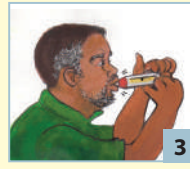
How to measure peak expiratory flow rate (PEFR)



Move marker to bottom of numbered scale.



- Stand up and take a full, deep breath.
- Hold breath and place mouthpiece between teeth.
- Form a seal with lips.



Breathe out as hard and as fast as possible (keeping fingers clear of scale).



- Read the result.
- Move marker back to bottom and repeat twice. Use the highest of the 3 readings.

How to assess response to inhaled beta-agonist

Calculate % PEFR response to inhaled beta-agonist to help diagnose asthma

- Measure 'initial PEFR'. Use the highest reading of 3 results.
- Give inhaled **salbutamol** 200mcg (2 puffs via a spacer) and wait for 15 minutes.
- Repeat PEFR - this is the 'repeat PEFR'
- Calculate % PEFR response = $\frac{(\text{repeat PEFR} - \text{initial PEFR})}{\text{Initial PEFR}} \times 100$
- If % PEFR response is $\geq 20\%$, **asthma** likely.

Using inhalers and spacers²

- If patient unable to use an inhaler correctly, add a spacer to increase drug delivery to lungs, especially if using inhaled corticosteroids. This may also reduce the risk of oral candida.
- Clean the spacer before first use and weekly: remove the canister and wash spacer with soapy water. Allow to drip dry. Avoid rinsing with water after each use.



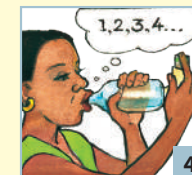
Shake inhaler and insert into spacer.



Stand up and breathe out. Then form a seal with lips around mouthpiece.



Press pump once to release one puff into spacer.



- Then take 4 breaths keeping spacer in mouth.
- Repeat step 3 and 4 for each puff.
- Rinse mouth after using inhaled corticosteroid.

¹Peak expiratory flow rate. ²If no spacer available, explain how to use inhaler without spacer: take off cap and shake inhaler. Stand up and breathe out. Then form seal with lips around inhaler mouthpiece. Breathe in slowly. As breathing in, press pump once and keep breathing in slowly. Close mouth and hold breath for 10 seconds. Breathe out.